Fiscal Year:	FY 2015	**Task Last Updated:**	FY 05/24/2016
PI Name:	Kim, Jungho Ph.D.		
Project Title:	Measurement and Analysis of Gravity Effects on Flow Boiling Heat Transfer Using IR Thermometry		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
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No. of PhD Candidates:		**No. of Master' Degrees:**	
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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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	EDITOR'S NOTE (5/16/2016): Continuation of "Development and Validation of Design Tools for Advanced, Two Phase, Space Heat Exchangers" with same Principal Investigator Dr. Jungho Kim, grant #NNX09AK39A. Added to Task Book when received information. The ability of flow boiling to provide high heat transfer coefficient over a small temperature range makes it suitable for many cooling applications. The experiment is one of numerous ongoing efforts by researchers in the US, Japan, and Europe seeking to resolve the complex issues during flow boiling in low-gravity environments. If such issues can be resolved, the size and weight of heat exchangers can be significantly reduced, allowing additional payload capability to be accommodated. The primary objective of the experiment is to understand the major mechanisms of heat transfer during flow boiling in a reduced gravity environment. Better understanding of different mechanisms and the quantification of parametric effects will be utilized to achieve the required heat fluxes for applications like electronic cooling and design of space-based hardware.		

Task Description:	The data and models needed to predict the behavior of two-phase flows in geometries relevant to advanced heat exchangers in variable gravity environments are being developed. Unlike previous work where only time- or space-averaged heat data were measured, we have obtained local measurements of the wall heat transfer coefficients using a novel IR (infrared) thermometry technique. An IR camera was used to visualize fluid flow within a silicon tube and obtain temperature distributions along the inside and outside walls of the tube so local heat transfer coefficients could be determined. The data is being used to develop and validate sub-models of the flow within each flow regime. Understanding of the heat transfer mechanisms will enable space-based heat exchangers to be designed with much more confidence. Although this work is primarily aimed at thermal management subsystems and development of lightweight evaporators, it also impacts other systems for novel power and transmission, advanced chemical propulsion, and energy storage.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The data is being used to develop and validate sub-models of the flow within each flow regime. Understanding of the heat transfer mechanisms will enable space-based heat exchangers to be designed with much more confidence. Although this work is primarily aimed at thermal management subsystems and development of lightweight evaporators, it also impacts other systems for novel power and transmission, advanced chemical propulsion, and energy storage.
Task Progress:	Year 1 Progress: The work over year 1 focused on modification of the test apparatus used to obtain low-g heat transfer. The primary modification was the addition of void fraction probes before and after the test section so the two-phase pressure drop could be determined with accuracy. The existing test apparatus was modified so that void fraction measurement before and after the test section could be measured using capacitance probes. This work was performed in conjunction with the group of Prof. Catherine Colin of Toulouse University, France. Year 2 Progress: The test apparatus, including the void fraction sensor provided by Prof. Catherine Colin of Toulouse University was flown on a low-g aircraft platform in Bordeaux, France in March, 2013. Heat transfer coefficient distributions along a 6 mm ID tube were obtained at various mass flow rates and wall heat fluxes. Data analysis (heat transfer+void fraction) was performed using this data. Specifically, measurements collected using the capacitance void fraction probe allowed the analysis of flow patterns and behavior in variable gravity conditions. Year 3 Progress: Slug flow, characterized by elongated bubbles separated by liquid slugs, is frequently seen in pulsating heat pipes, microchannels, and in macro-scale channels at low vapor qualities. Vapor/gas slugs rising in a vertical column are called Taylor bubbles. To characterize the heat transfer and dynamics of rising Taylor bubbles, a flow boiling experiment was conducted in which measurements of the local wall heat transfer and film thickness along with high speed images were obtained as single bubbles of varying length rose in a vertical column containing upward flowing liquid. The study of single bubbles was chosen in lieu of bubble trains so flow conditions upstream and downstream of the bubbles could be measured, reducing the complexity in approximating the flow patterns and understanding the heat transfer profile around each bubble. The flow loop was designed to create and release single Taylor bubbles. This section consisted of a bubble generation segment and a bypass segment connected at the downstream end by a three-way valve. To generate a bubble, the valve was set to divert the liquid flow through the bypass segment, while a wire heater evaporated liquid in the bubble generation segment. The bubble volume was varied by adjusting the power to the wire and the heating time. Once the desired bubble volume was generated, it was released by rotating the valve such that liquid was redirected through the bubble generation segment, pushing the bubble into the test section. The bubble rose vertically into a 6 mm ID silicon test section positioned 200 mm downstream of the three-way valve where heat transfer measurements and flow visualization were made. Pressure taps were located at the inlet and outlet of silicon tube so differential and absolute pressures could be measured. The absolute pressure transducer was used to determine the saturation temperature of the fluid entering the test section. Immediately after leaving the heated silicon tube, the rising bubble passed through a glass adiabatic section (380 mm downstream of the three-way valve) where high speed video was obtained using a CMOS video camera at frame rates between 900 and 1200 frames per second. The high-speed visualization was used to determine the bubble length, study the dynamics of the tail, and analyze the bubble shape. A laser displacement sensor was used to measure the liquid film thickness around the bubble. The wall heat transfer was measured for each bubble as they passed through the test section. Ahead of the bubble nose, the heat transfer coefficient corresponded to single-phase flow. As the bubble passed, the flow as forced into the liquid film where it accelerated, thinning the thermal boundary layer and increasing the heat transfer coefficient slightly. Vortices were generated when the downward moving film passed the bubble tail and interacted with the trailing liquid slug, inducing turbulence and a large spike in heat transfer. Once the turbulent mixing behind the bubble died out, a thermally developing boundary layer was re-established and the heat transfer coefficient decayed to the single-phase flow value. The largest contribution to heat transfer enhancement over single-phase flow occurred in the wake region immediately behind the bubble, a finding in opposition to current slug flow models which assume the highest heat transfer occurs in the evaporating thin liquid film. The test apparatus was modified so it could be flown on a low-g aircraft. Tests were conducted on NASA’s C-9 aircraft between June 22-26, 2015 from Ellington Field to obtain heat transfer data for Taylor bubbles in low-g, high-g, and a few Martian-g environments. The data is currently being processed and we expect to submit the results to a journal in a couple of months. EDITOR'S NOTE (5/16/2016): Continued by "Gravity Effects on Flow Boiling Heat Transfer Using Local Heat Transfer Distribution Methods" with same Principal Investigator Dr. Jungho Kim, grant #NNX15AK16G. See that project for subsequent reporting.
Bibliography Type:	Description: (Last Updated: 03/13/2026)

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